

PMC Compass: Modelling Outputs for SLE

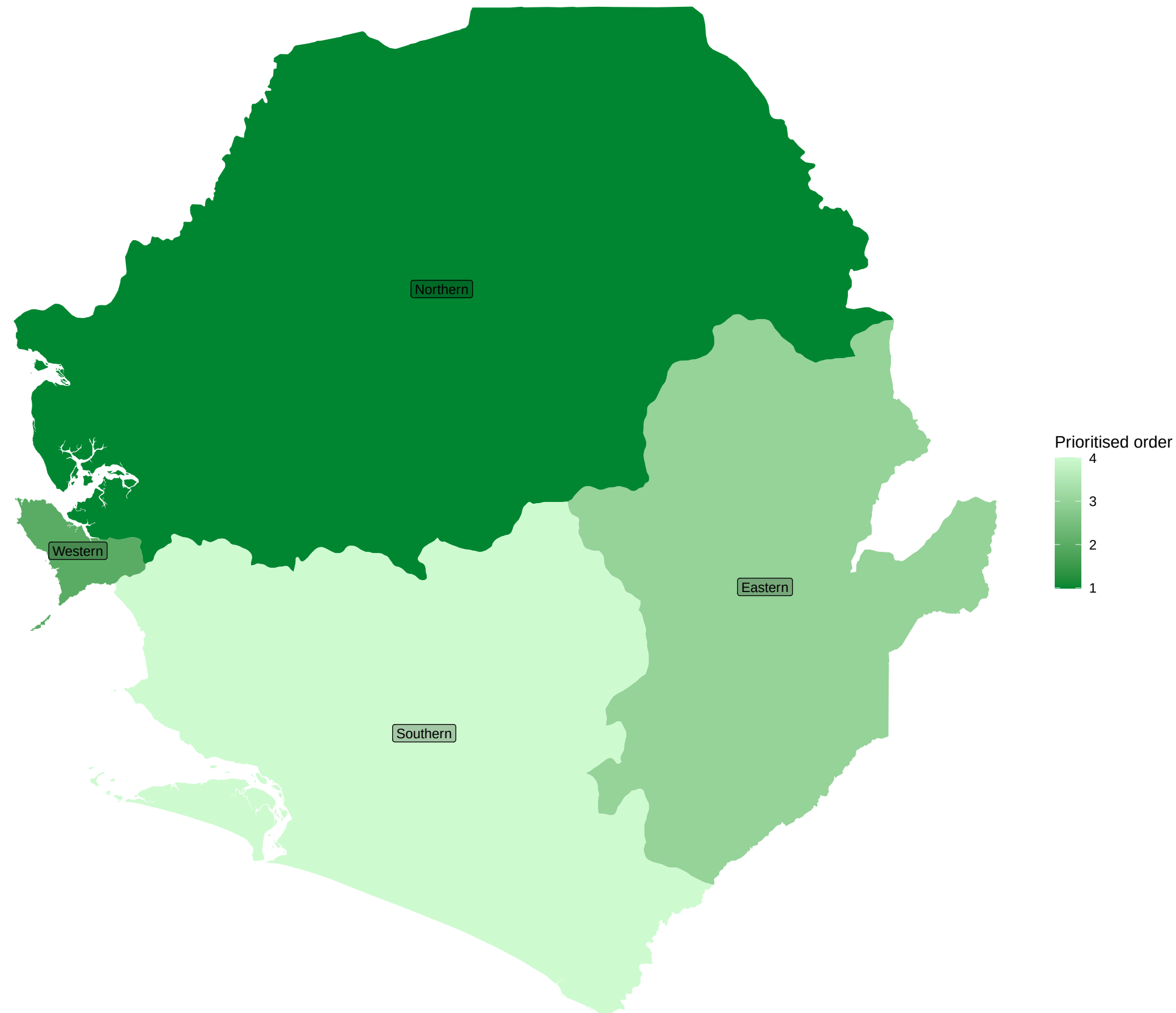
Modelling outputs from PMC Compass. Please note, results are preliminary and subject to change following further validation.

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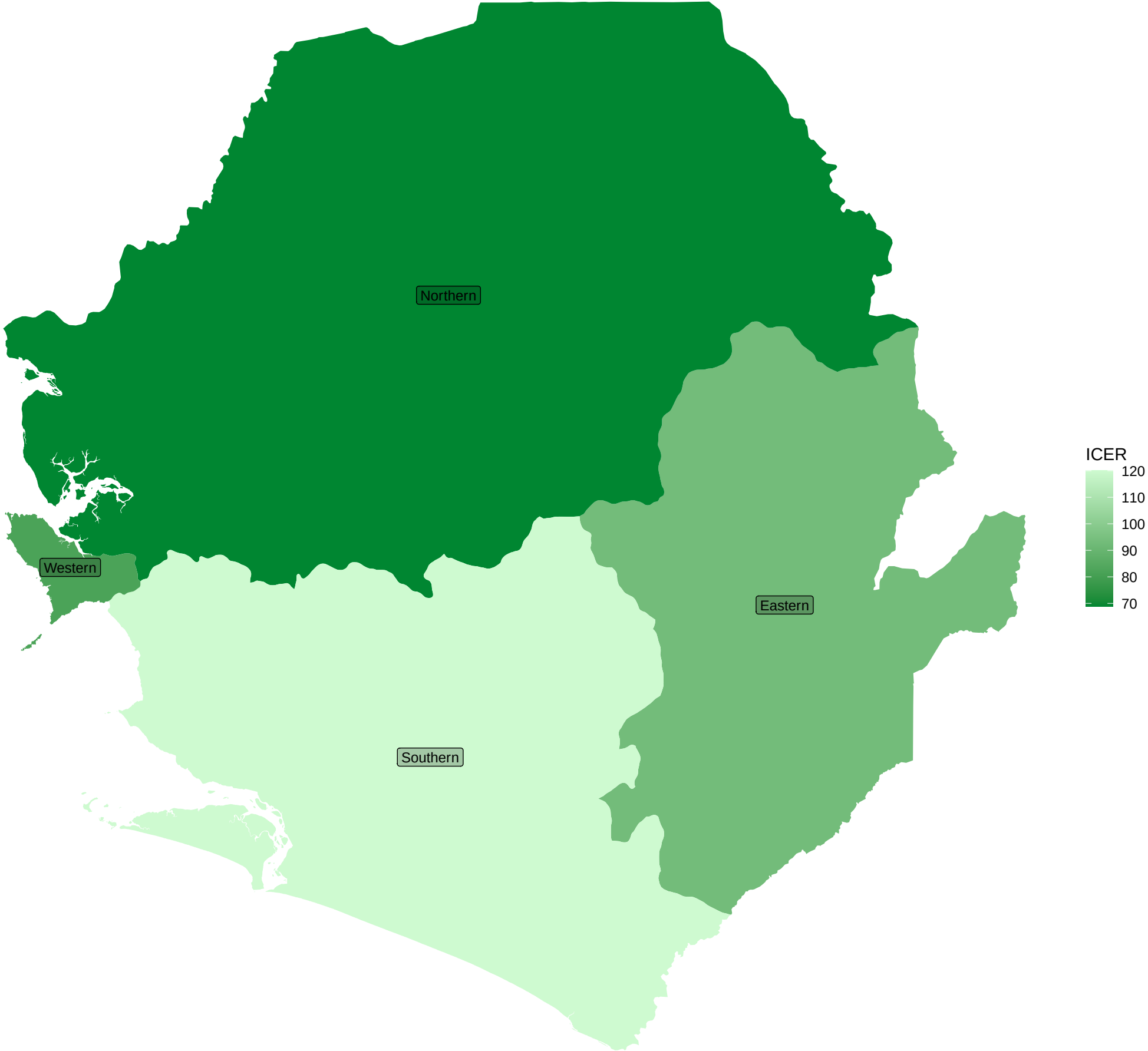
SLE: Ranked prioritised order to implement PMC

Order at which to implement PMC in the country based on cost-effectiveness of the PMC delivery strategy.



SLE: Incremental cost-effectiveness ratios of implementing PMC co-delivery with EPI only vs No PMC in each admin-1 unit.

Incremental cost-effectiveness ratios of implementing PMC co-delivery with EPI only vs No PMC in each admin-1 unit. All costs used to calculate these exclude national-level start up costs.



SLE: Ranked prioritisation of options (page 1)

The first row indicates the single region and delivery model where PMC would be most cost-effective, meaning it would gain the most health (in disability-adjusted life-years) for the resources invested (public health service + donor costs of PMC and malaria treatment). The second row indicates the next-most cost-effective option to consider implementing alongside the first option, if additional implementation resources remain.

Country	Region (admin-1)	PMC delivery strategy (vs. comparator)	Additional clinical malaria cases averted	Additional hospitalisations averted	Additional deaths averted	Additional DALYs averted
SLE	Northern	PMC co-delivery with EPI only vs No PMC	31113	1133	244	1981
SLE	Western	PMC co-delivery with EPI only vs No PMC	9321	502	108	650
SLE	Eastern	PMC co-delivery with EPI only vs No PMC	15672	653	141	1028
SLE	Southern	PMC co-delivery with EPI only vs No PMC	16249	713	154	1077

SLE: Ranked prioritisation of options (page 2)

The first row indicates the single region and delivery model where PMC would be most cost-effective, meaning it would gain the most health (in disability-adjusted life-years) for the resources invested (public health service + donor costs of PMC and malaria treatment). The second row indicates the next-most cost-effective option to consider implementing alongside the first option, if additional implementation resources remain.

Country	Region (admin-1)	PMC delivery strategy (vs. comparator)	Financial cost of SP and administration consumables	Cumulative cost of SP and administration consumables	Additional economic cost savings to public providers from reduced treatment	Cumulative economic cost savings to public providers from reduced treatment
SLE	Northern	PMC co-delivery with EPI only vs No PMC	71470	71470	-119972	-119972
SLE	Western	PMC co-delivery with EPI only vs No PMC	51226	122696	-45367	-165339
SLE	Eastern	PMC co-delivery with EPI only vs No PMC	50639	173335	-65200	-230539
SLE	Southern	PMC co-delivery with EPI only vs No PMC	53692	227027	-69684	-300223

SLE: Ranked prioritisation of options (page 3)

The first row indicates the single region and delivery model where PMC would be most cost-effective, meaning it would gain the most health (in disability-adjusted life-years) for the resources invested (public health service + donor costs of PMC and malaria treatment). The second row indicates the next-most cost-effective option to consider implementing alongside the first option, if additional implementation resources remain.

Country	Region (admin-1)	PMC delivery strategy (vs. comparator)	Additional net economic costs of PMC implementation and treatment	Cumulative net costs of PMC implementation and treatment	Incremental economic cost per DALY averted vs. next higher option in ranking	Cost-effective? (lower country-specific threshold)
SLE	Northern	PMC co-delivery with EPI only vs No PMC	282947	282947	143	no
SLE	Western	PMC co-delivery with EPI only vs No PMC	53119	336066	82	yes
SLE	Eastern	PMC co-delivery with EPI only vs No PMC	95131	431197	93	no
SLE	Southern	PMC co-delivery with EPI only vs No PMC	129604	560801	120	no

SLE: Ranked prioritisation of options (page 4)

The first row indicates the single region and delivery model where PMC would be most cost-effective, meaning it would gain the most health (in disability-adjusted life-years) for the resources invested (public health service + donor costs of PMC and malaria treatment). The second row indicates the next-most cost-effective option to consider implementing alongside the first option, if additional implementation resources remain.

Country	Region (admin-1)	PMC delivery strategy (vs. comparator)	Cost-effective? (higher country-specific threshold)	Cost-effective? (\$250 / DALY averted threshold)	Population (all ages)	Expected PMC doses delivered (age 0-1yr)
SLE	Northern	PMC co-delivery with EPI only vs No PMC	yes	yes	2937401	370275
SLE	Western	PMC co-delivery with EPI only vs No PMC	yes	yes	1992429	264472
SLE	Eastern	PMC co-delivery with EPI only vs No PMC	yes	yes	1926376	259045
SLE	Southern	PMC co-delivery with EPI only vs No PMC	yes	yes	2027967	275756

SLE: Ranked prioritisation of options (page 5)

The first row indicates the single region and delivery model where PMC would be most cost-effective, meaning it would gain the most health (in disability-adjusted life-years) for the resources invested (public health service + donor costs of PMC and malaria treatment). The second row indicates the next-most cost-effective option to consider implementing alongside the first option, if additional implementation resources remain.

Country	Region (admin-1)	PMC delivery strategy (vs. comparator)	Expected PMC doses delivered (age 1-2yr)	Expected PMC doses delivered (age 2-2.5yr)	Data source for cost modelling	ICER based on regional costs only (no national costs)
SLE	Northern	PMC co-delivery with EPI only vs No PMC	45281	0	Plus Project countries	69
SLE	Western	PMC co-delivery with EPI only vs No PMC	33000	0	Plus Project countries	82
SLE	Eastern	PMC co-delivery with EPI only vs No PMC	34043	0	Plus Project countries	93
SLE	Southern	PMC co-delivery with EPI only vs No PMC	35450	0	Plus Project countries	120

SLE: Ranked prioritisation of options (page 6)

The first row indicates the single region and delivery model where PMC would be most cost-effective, meaning it would gain the most health (in disability-adjusted life-years) for the resources invested (public health service + donor costs of PMC and malaria treatment). The second row indicates the next-most cost-effective option to consider implementing alongside the first option, if additional implementation resources remain.

Country	Region (admin-1)	PMC delivery strategy (vs. comparator)	rank
SLE	Northern	PMC co-delivery with EPI only vs No PMC	1
SLE	Western	PMC co-delivery with EPI only vs No PMC	2
SLE	Eastern	PMC co-delivery with EPI only vs No PMC	3
SLE	Southern	PMC co-delivery with EPI only vs No PMC	4

SLE: Cost assumptions (page 1)

Detailed information on the assumptions used in estimating costs, DALYs, and cost–effectiveness used in the main ranking of options.

Iso code	Region (admin–1 unit)	Population (all ages)	Scenario	N doses in PMC schedule	Data source for cost modelling	Total cost of PMC implementation (excl national costs)
SLE	Eastern	1926376	No PMC	0	Plus Project countries	0
SLE	Eastern	1926376	PMC co–delivery with EPI only	6	Plus Project countries	160331
SLE	Northern	2937401	No PMC	0	Plus Project countries	0
SLE	Northern	2937401	PMC co–delivery with EPI only	6	Plus Project countries	257357
SLE	Southern	2027967	No PMC	0	Plus Project countries	0
SLE	Southern	2027967	PMC co–delivery with EPI only	6	Plus Project countries	199288
SLE	Western	1992429	No PMC	0	Plus Project countries	0
SLE	Western	1992429	PMC co–delivery with EPI only	6	Plus Project countries	98486

SLE: Cost assumptions (page 2)

Detailed information on the assumptions used in estimating costs, DALYs, and cost–effectiveness used in the main ranking of options.

Iso code	Region (admin–1 unit)	Population (all ages)	Scenario	Sub–total: Cost of treatment	Total cost of PMC implementation + treatment (excluding national costs)	Total cost of PMC implementation + treatment (including full national costs for every region)
SLE	Eastern	1926376	No PMC	635233	635233	635233
SLE	Eastern	1926376	PMC co–delivery with EPI only	570033	730364	875926
SLE	Northern	2937401	No PMC	1249871	1249871	1249871
SLE	Northern	2937401	PMC co–delivery with EPI only	1129899	1387256	1532818
SLE	Southern	2027967	No PMC	663182	663182	663182
SLE	Southern	2027967	PMC co–delivery with EPI only	593498	792786	938348
SLE	Western	1992429	No PMC	464298	464298	464298
SLE	Western	1992429	PMC co–delivery with EPI only	418931	517417	662979

SLE: Cost assumptions (page 3)

Detailed information on the assumptions used in estimating costs, DALYs, and cost–effectiveness used in the main ranking of options.

Iso code	Region (admin–1 unit)	Population (all ages)	Scenario	Clinical cases (N, 0–30mo)	Clinical cases for whom care sought	Hospitalisations (N, 0–30mo)
SLE	Eastern	1926376	No PMC	179896	89948	5543
SLE	Eastern	1926376	PMC co–delivery with EPI only	164224	82112	4890
SLE	Northern	2937401	No PMC	393399	196700	9719
SLE	Northern	2937401	PMC co–delivery with EPI only	362286	181143	8586
SLE	Southern	2027967	No PMC	185117	92558	5868
SLE	Southern	2027967	PMC co–delivery with EPI only	168868	84434	5155
SLE	Western	1992429	No PMC	107750	53875	4766
SLE	Western	1992429	PMC co–delivery with EPI only	98429	49214	4264

SLE: Cost assumptions (page 4)

Detailed information on the assumptions used in estimating costs, DALYs, and cost–effectiveness used in the main ranking of options.

Iso code	Region (admin–1 unit)	Population (all ages)	Scenario	Deaths (N, 0–30mo)	Total DALYs	DALYs from clinical cases
SLE	Eastern	1926376	No PMC	1192	11104	9175
SLE	Eastern	1926376	PMC co–delivery with EPI only	1051	10076	8375
SLE	Northern	2937401	No PMC	2090	23446	20063
SLE	Northern	2937401	PMC co–delivery with EPI only	1846	21465	18477
SLE	Southern	2027967	No PMC	1262	11483	9441
SLE	Southern	2027967	PMC co–delivery with EPI only	1108	10406	8612
SLE	Western	1992429	No PMC	1025	7154	5495
SLE	Western	1992429	PMC co–delivery with EPI only	917	6504	5020

SLE: Cost assumptions (page 5)

Detailed information on the assumptions used in estimating costs, DALYs, and cost–effectiveness used in the main ranking of options.

Iso code	Region (admin–1 unit)	Population (all ages)	Scenario	DALYs from hospitalisations	DALYs from deaths	PMC cost input: Implementation cost at national level
SLE	Eastern	1926376	No PMC	737	1192	0
SLE	Eastern	1926376	PMC co–delivery with EPI only	650	1051	145562
SLE	Northern	2937401	No PMC	1293	2090	0
SLE	Northern	2937401	PMC co–delivery with EPI only	1142	1846	145562
SLE	Southern	2027967	No PMC	780	1262	0
SLE	Southern	2027967	PMC co–delivery with EPI only	686	1108	145562
SLE	Western	1992429	No PMC	634	1025	0
SLE	Western	1992429	PMC co–delivery with EPI only	567	917	145562

SLE: Cost assumptions (page 6)

Detailed information on the assumptions used in estimating costs, DALYs, and cost–effectiveness used in the main ranking of options.

Iso code	Region (admin–1 unit)	Population (all ages)	Scenario	PMC cost input: PMC implementation cost per district	PMC cost input: PMC implementation cost per health facility	PMC cost input: Cost of PMC administration consumables per dose
SLE	Eastern	1926376	No PMC	0	0	0.00
SLE	Eastern	1926376	PMC co–delivery with EPI only	25673	174	0.05
SLE	Northern	2937401	No PMC	0	0	0.00
SLE	Northern	2937401	PMC co–delivery with EPI only	25673	174	0.05
SLE	Southern	2027967	No PMC	0	0	0.00
SLE	Southern	2027967	PMC co–delivery with EPI only	25673	174	0.05
SLE	Western	1992429	No PMC	0	0	0.00
SLE	Western	1992429	PMC co–delivery with EPI only	25673	174	0.05

SLE: Cost assumptions (page 7)

Detailed information on the assumptions used in estimating costs, DALYs, and cost–effectiveness used in the main ranking of options.

Iso code	Region (admin–1 unit)	Population (all ages)	Scenario	PMC Cost input: Cost per SP tablet (includes wastage)	N districts in region	N health facilities in region
SLE	Eastern	1926376	No PMC	0.00	3	272
SLE	Eastern	1926376	PMC co–delivery with EPI only	0.11	3	272
SLE	Northern	2937401	No PMC	0.00	5	450
SLE	Northern	2937401	PMC co–delivery with EPI only	0.11	5	450
SLE	Southern	2027967	No PMC	0.00	4	336
SLE	Southern	2027967	PMC co–delivery with EPI only	0.11	4	336
SLE	Western	1992429	No PMC	0.00	2	62
SLE	Western	1992429	PMC co–delivery with EPI only	0.11	2	62

SLE: Cost assumptions (page 8)

Detailed information on the assumptions used in estimating costs, DALYs, and cost–effectiveness used in the main ranking of options.

Iso code	Region (admin–1 unit)	Population (all ages)	Scenario	N expected PMC doses delivered (age 0–1yr)	N expected PMC doses delivered (age 1–2yr)	N expected PMC doses delivered (age 2–2.5yr)
SLE	Eastern	1926376	No PMC	0	0	0
SLE	Eastern	1926376	PMC co–delivery with EPI only	259045	34043	0
SLE	Northern	2937401	No PMC	0	0	0
SLE	Northern	2937401	PMC co–delivery with EPI only	370275	45281	0
SLE	Southern	2027967	No PMC	0	0	0
SLE	Southern	2027967	PMC co–delivery with EPI only	275756	35450	0
SLE	Western	1992429	No PMC	0	0	0
SLE	Western	1992429	PMC co–delivery with EPI only	264472	33000	0

SLE: Cost assumptions (page 9)

Detailed information on the assumptions used in estimating costs, DALYs, and cost–effectiveness used in the main ranking of options.

Iso code	Region (admin–1 unit)	Population (all ages)	Scenario	N total expected PMC doses delivered (age 0–2.5yr)	N SP tablets delivered	Sub–total: Cost of PMC implementation at district and HFs
SLE	Eastern	1926376	No PMC	0	0	0
SLE	Eastern	1926376	PMC co–delivery with EPI only	293088	327131	124347
SLE	Northern	2937401	No PMC	0	0	0
SLE	Northern	2937401	PMC co–delivery with EPI only	415556	460837	206665
SLE	Southern	2027967	No PMC	0	0	0
SLE	Southern	2027967	PMC co–delivery with EPI only	311206	346656	161156
SLE	Western	1992429	No PMC	0	0	0
SLE	Western	1992429	PMC co–delivery with EPI only	297473	330473	62134

SLE: Cost assumptions (page 10)

Detailed information on the assumptions used in estimating costs, DALYs, and cost–effectiveness used in the main ranking of options.

Iso code	Region (admin–1 unit)	Population (all ages)	Scenario	Sub–total: Cost of PMC administration (SP+consumables)	Treatment cost per clinical case	Treatment cost per hospitalisation
SLE	Eastern	1926376	No PMC	0	3.49	47.71
SLE	Eastern	1926376	PMC co–delivery with EPI only	50639	3.49	47.71
SLE	Northern	2937401	No PMC	0	3.49	47.71
SLE	Northern	2937401	PMC co–delivery with EPI only	71470	3.49	47.71
SLE	Southern	2027967	No PMC	0	3.49	47.71
SLE	Southern	2027967	PMC co–delivery with EPI only	53692	3.49	47.71
SLE	Western	1992429	No PMC	0	3.49	47.71
SLE	Western	1992429	PMC co–delivery with EPI only	51226	3.49	47.71

SLE: Cost assumptions (page 11)

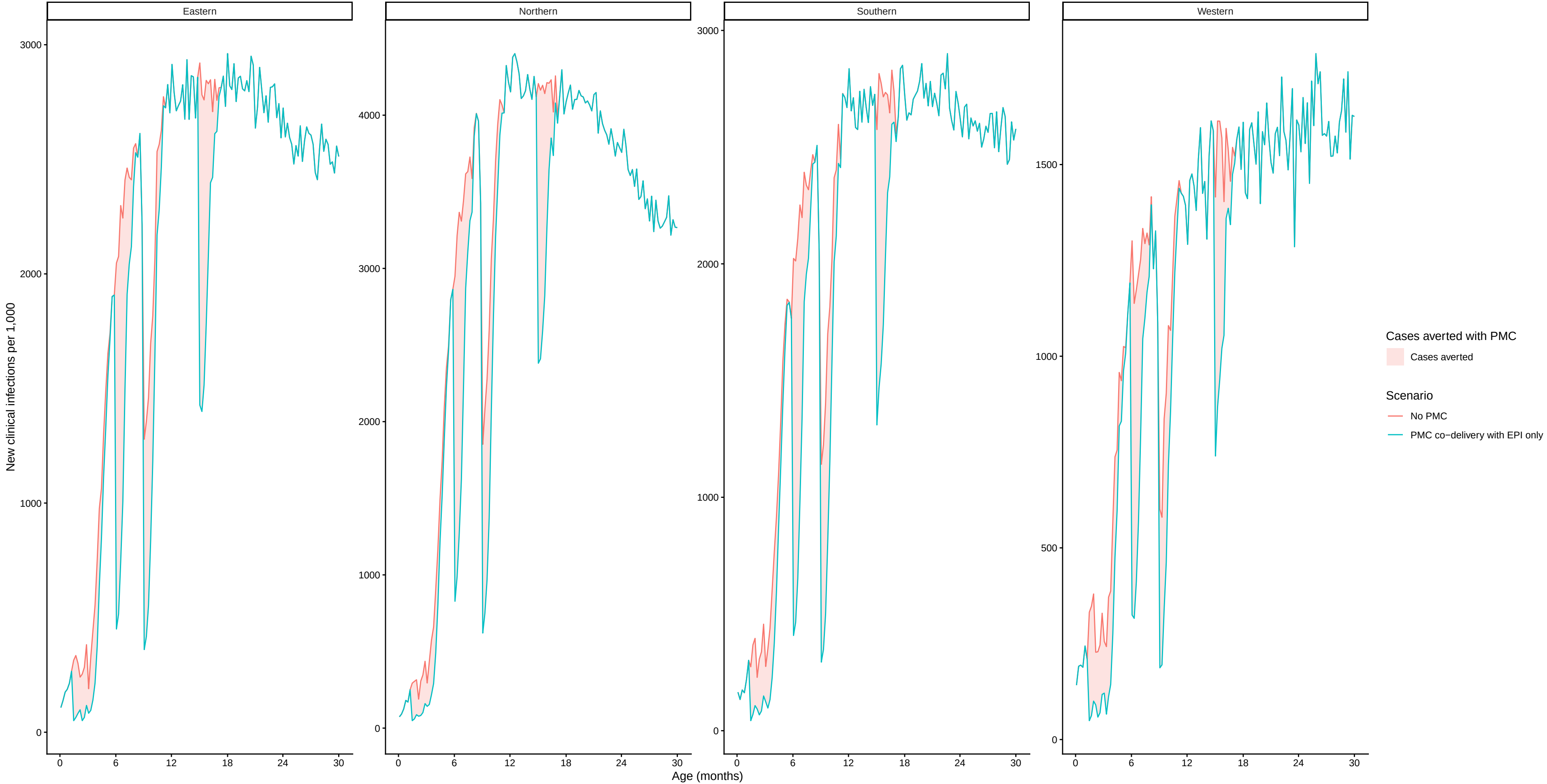
Detailed information on the assumptions used in estimating costs, DALYs, and cost–effectiveness used in the main ranking of options.

Iso code	Region (admin–1 unit)	Population (all ages)	Scenario	Treatment cost per death
SLE	Eastern	1926376	No PMC	47.71
SLE	Eastern	1926376	PMC co–delivery with EPI only	47.71
SLE	Northern	2937401	No PMC	47.71
SLE	Northern	2937401	PMC co–delivery with EPI only	47.71
SLE	Southern	2027967	No PMC	47.71
SLE	Southern	2027967	PMC co–delivery with EPI only	47.71
SLE	Western	1992429	No PMC	47.71
SLE	Western	1992429	PMC co–delivery with EPI only	47.71

SLE: Clinical malaria incidence

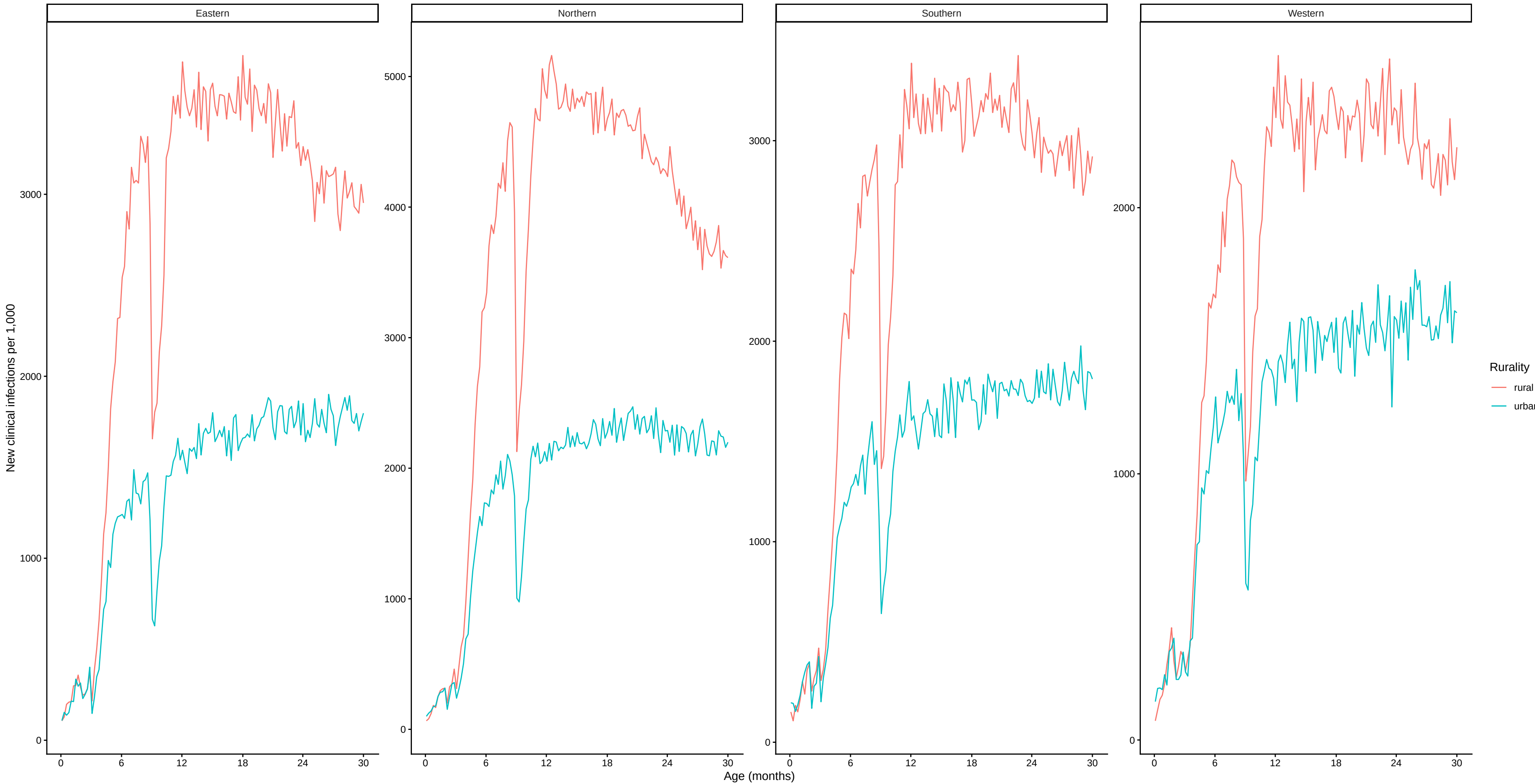
Age-specific clinical malaria cases per 1,000 children (0–30 months, with and without PMC).

PMC is assumed to be co-delivered with EPI and have a 90% uptake relative to the coverage of the co-delivered intervention.



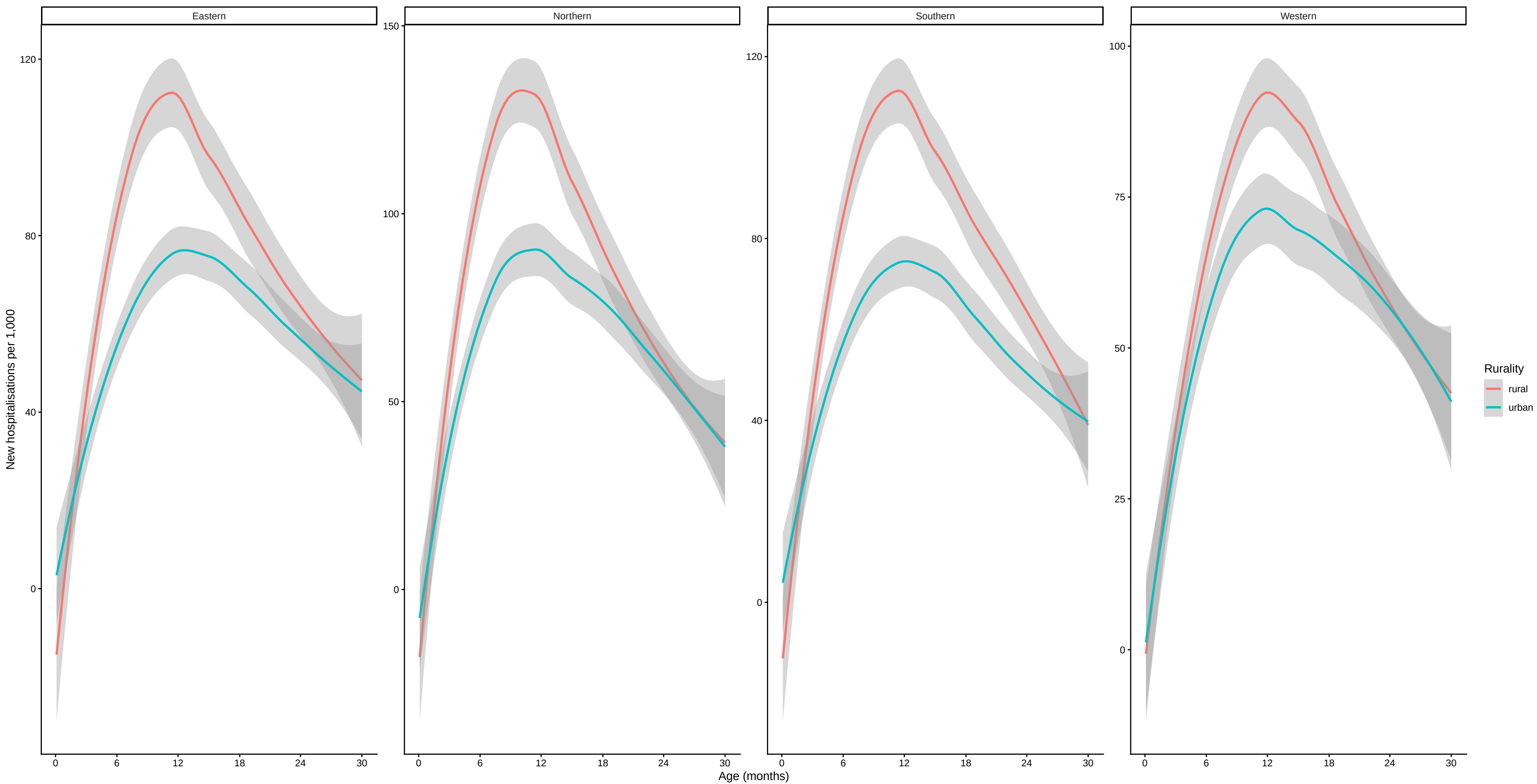
SLE: Clinical malaria incidence (rural and urban)

Age-specific clinical malaria cases per 1,000 children (0–30 months, no PMC), stratified by rural and urban residence.



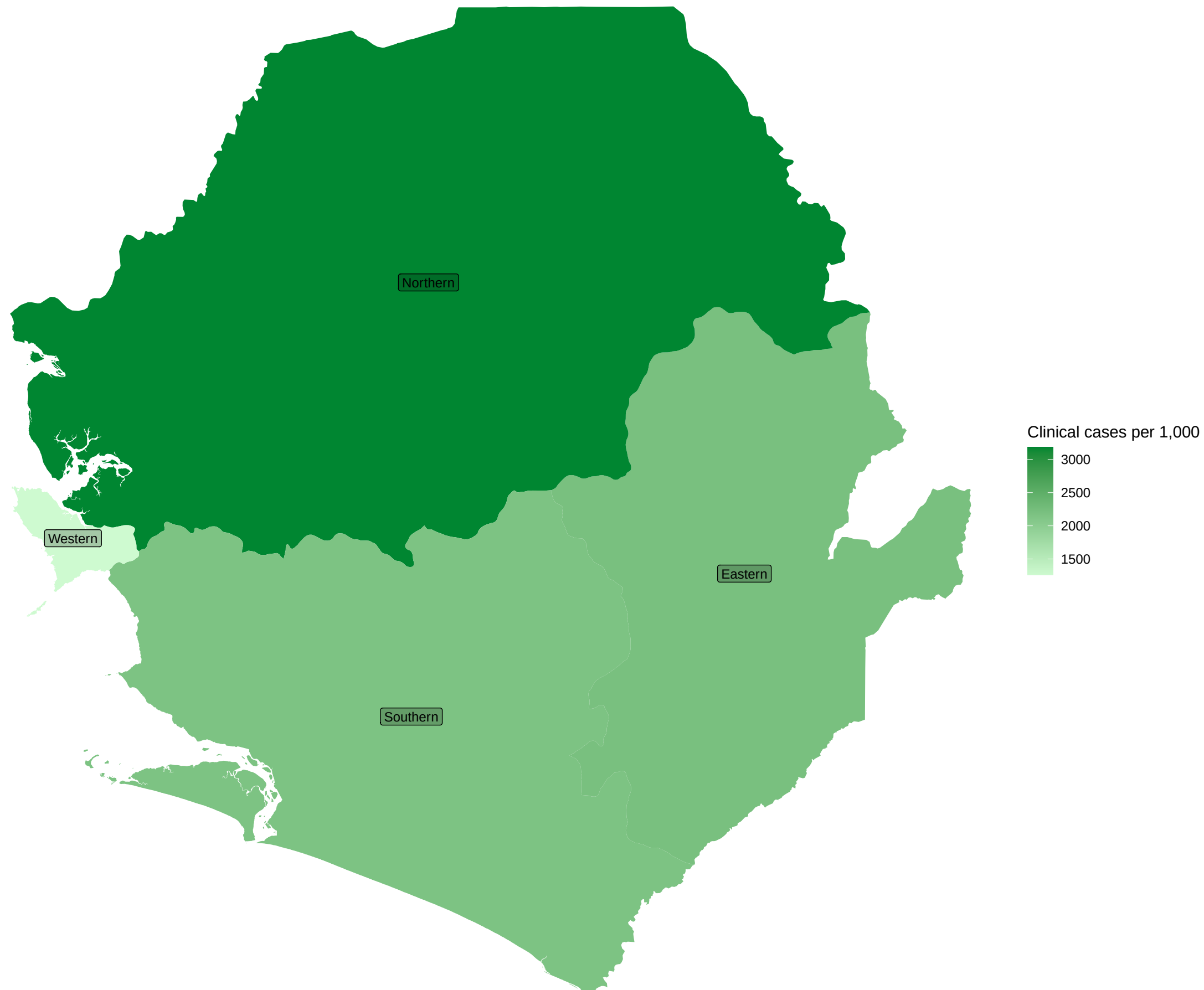
SLE: Malaria hospitalisations

Age-specific new malaria hospitalisations per 1,000 children (0–30 months, no PMC), stratified by rural and urban residence.
Raw model outputs have been smoothed to reduce stochastic variation.



SLE: Spatial distribution of clinical malaria without PMC

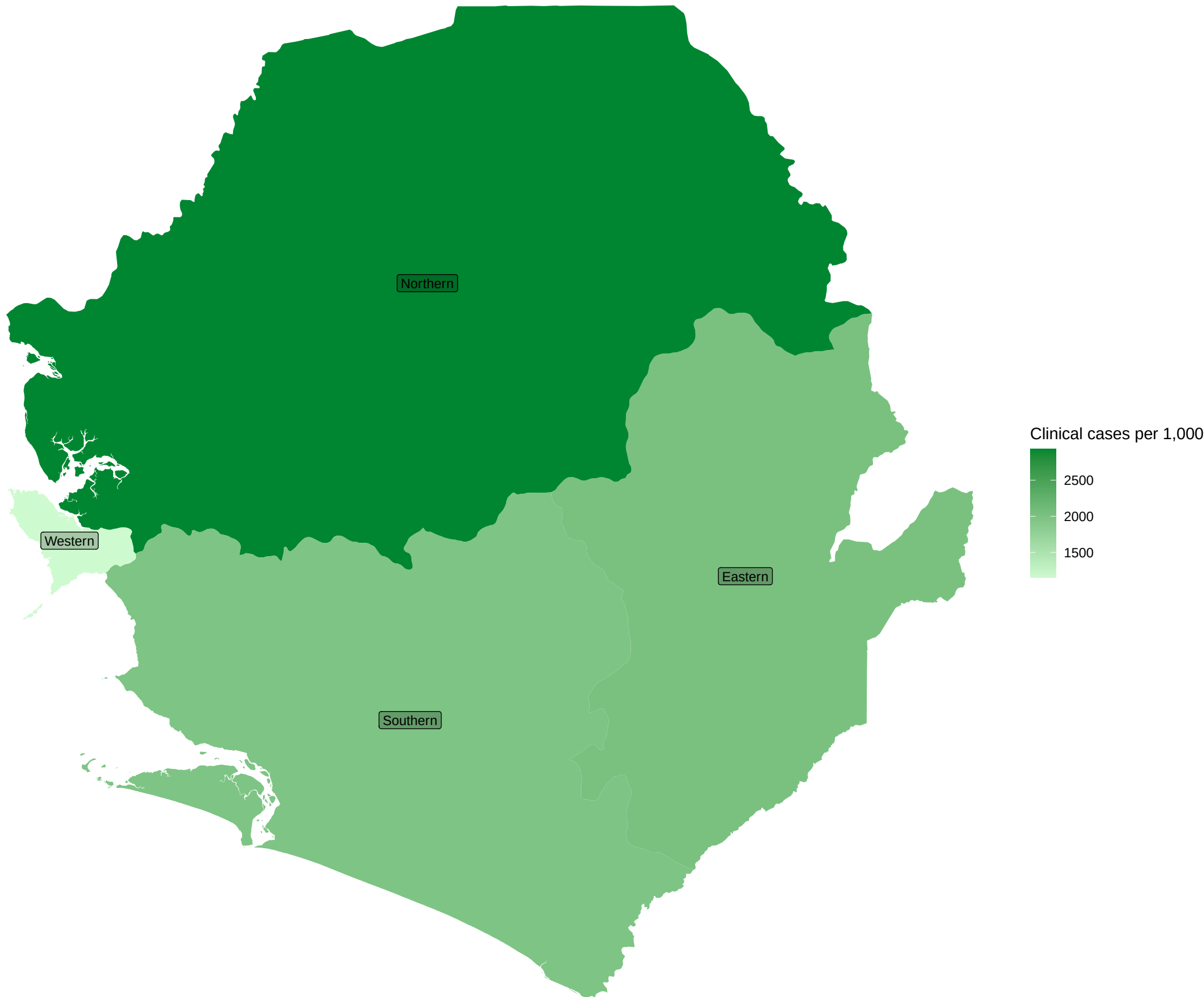
Admin-1 clinical malaria incidence per 1,000 children aged (0–30 months, no PMC).



SLE: Spatial distribution of clinical malaria with PMC

Admin-1 clinical malaria incidence per 1,000 children aged (0–30 months, with PMC).

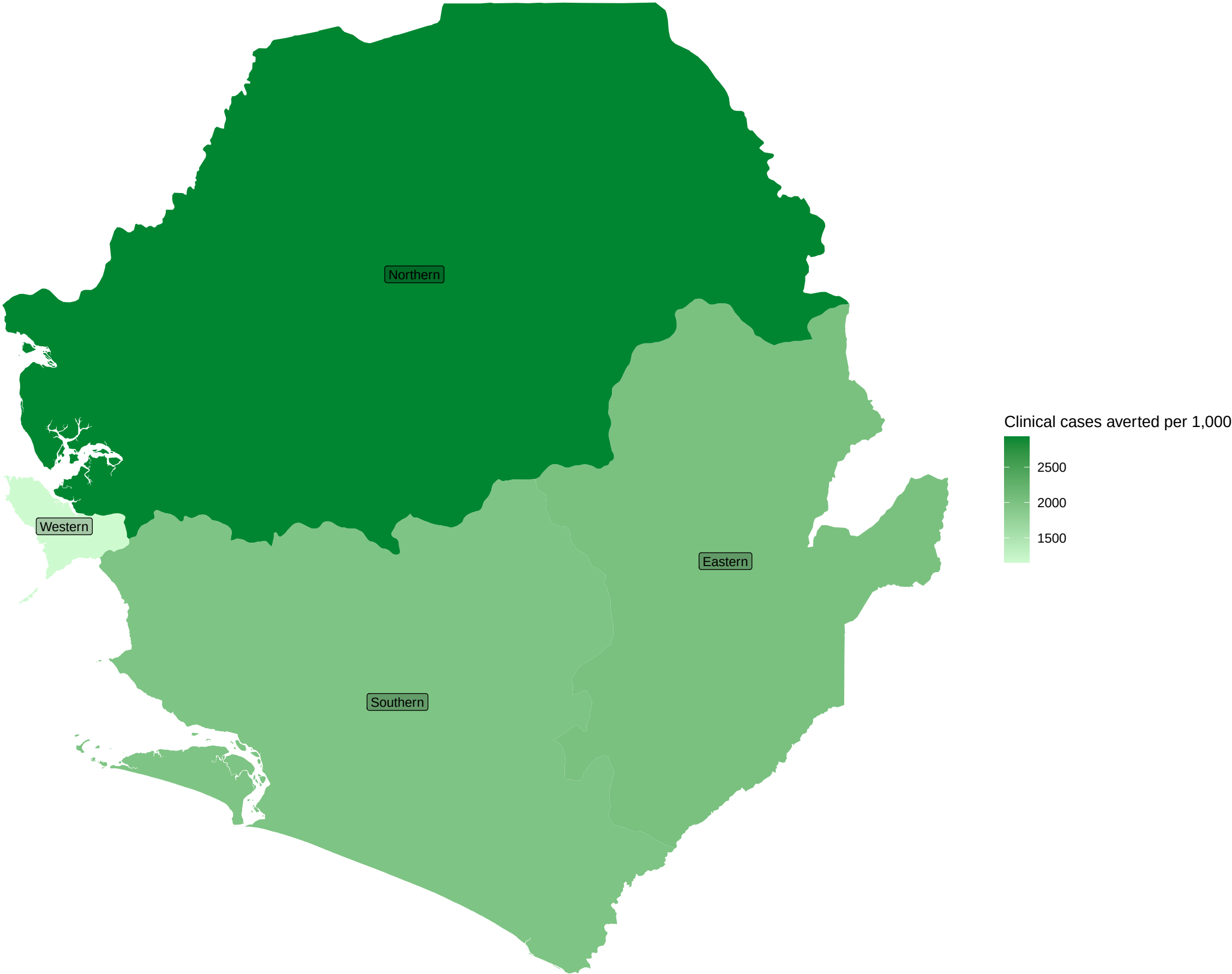
PMC is assumed to be co-delivered with EPI and have a 90% uptake relative to the coverage of the co-delivered intervention.



SLE: Spatial distribution of averted clinical malaria with PMC

Admin-1 averted clinical malaria incidence per 1,000 children aged (0–30 months, averted with PMC).

PMC is assumed to be co-delivered with EPI and have a 90% uptake relative to the coverage of the co-delivered intervention.



SLE: Clinical malaria burden (page 1)

Annual clinical malaria burden per 1,000 children aged 0–2.5 years, with and without PMC.

PMC is assumed to be co–delivered with EPI and have a 90% uptake relative to the coverage of the co–delivered intervention.

Admin–1	Clinical cases per 1,000, no PMC (0–2.5y)	Clinical cases per 1,000, with PMC (0–2.5y)	Clinical cases averted per 1,000 (0–2.5y)
Abia	362.1	325.8	36.3
Adamawa	379.8	355.7	24.1
Akwa_Ibom	445.1	413.0	32.1
Anambra	247.1	225.2	21.9
Bauchi	474.3	457.9	16.4
Bayelsa	578.6	539.8	38.8
Benue	369.2	343.5	25.8
Borno	1287.6	1233.4	54.3
Cross_River	527.1	486.7	40.4
Delta	333.0	308.1	24.8
Ebonyi	752.8	694.8	58.1
Edo	705.8	650.6	55.2
Ekiti	1165.1	1065.5	99.7
Enugu	380.5	348.1	32.4
Federal_Capital_Territory	323.3	297.5	25.7
Gombe	448.2	431.8	16.4
Imo	260.2	239.5	20.7
Jigawa	1614.4	1517.4	96.9
Kaduna	594.6	570.2	24.4
Kano	826.2	788.8	37.3
Katsina	2178.7	2115.4	63.2
Kebbi	3879.9	3744.2	135.7
Kogi	606.9	568.5	38.5
Kwara	372.3	348.6	23.7
Lagos	123.6	111.7	11.9
Nasarawa	1022.6	952.2	70.4
Niger	1011.6	970.3	41.3
Ogun	412.8	389.4	23.4
Ondo	1074.9	992.2	82.7
Osun	718.5	679.2	39.3

SLE: Clinical malaria burden (page 2)

Annual clinical malaria burden per 1,000 children aged 0–2.5 years, with and without PMC.

PMC is assumed to be co–delivered with EPI and have a 90% uptake relative to the coverage of the co–delivered intervention.

Admin–1	Clinical cases per 1,000, no PMC (0–2.5y)	Clinical cases per 1,000, with PMC (0–2.5y)	Clinical cases averted per 1,000 (0–2.5y)
Oyo	582.4	552.3	30.1
Plateau	786.5	733.0	53.5
Rivers	384.2	352.8	31.4
Sokoto	3696.7	3601.7	95.0
Taraba	563.2	540.4	22.8
Yobe	780.6	737.7	42.9
Zamfara	3789.7	3718.0	71.7

SLE: Severe malaria burden (page 1)

Annual severe malaria hospitalisations per 1,000 children aged 0–2.5 years, with and without PMC.

PMC is assumed to be co–delivered with EPI and have a 90% uptake relative to the coverage of the co–delivered intervention.

Admin–1	Hospitalisations per 1,000, no PMC (0–2.5y)	Hospitalisations per 1,000, with PMC (0–2.5y)	Hospitalisations averted per 1,000 (0–2.5y)
Abia	24.1	21.3	2.8
Adamawa	24.7	23.0	1.8
Akwa_Ibom	26.5	24.3	2.2
Anambra	16.9	15.3	1.7
Bauchi	30.9	29.7	1.2
Bayelsa	32.2	29.7	2.5
Benue	23.2	21.5	1.7
Borno	59.7	56.8	2.9
Cross_River	29.5	26.8	2.7
Delta	21.8	19.9	1.8
Ebonyi	38.3	34.9	3.4
Edo	36.8	33.1	3.7
Ekiti	48.1	42.8	5.4
Enugu	24.7	22.4	2.3
Federal_Capital_Territory	20.4	18.8	1.7
Gombe	27.8	26.6	1.2
Imo	18.4	16.8	1.6
Jigawa	72.6	67.3	5.3
Kaduna	33.2	31.6	1.6
Kano	46.5	44.1	2.3
Katsina	84.0	81.1	2.9
Kebbi	100.5	95.2	5.4
Kogi	33.3	30.6	2.7
Kwara	22.0	20.4	1.6
Lagos	10.4	9.3	1.1
Nasarawa	43.4	39.6	3.8
Niger	43.1	40.7	2.4
Ogun	25.7	23.8	1.9
Ondo	45.6	41.0	4.6
Osun	35.3	32.8	2.5

SLE: Severe malaria burden (page 2)

Annual severe malaria hospitalisations per 1,000 children aged 0–2.5 years, with and without PMC.

PMC is assumed to be co–delivered with EPI and have a 90% uptake relative to the coverage of the co–delivered intervention.

Admin–1	Hospitalisations per 1,000, no PMC (0–2.5y)	Hospitalisations per 1,000, with PMC (0–2.5y)	Hospitalisations averted per 1,000 (0–2.5y)
Oyo	30.9	28.7	2.2
Plateau	39.2	35.9	3.2
Rivers	25.4	22.7	2.6
Sokoto	103.4	99.6	3.8
Taraba	32.3	30.7	1.6
Yobe	46.1	43.2	2.9
Zamfara	102.3	99.4	2.9